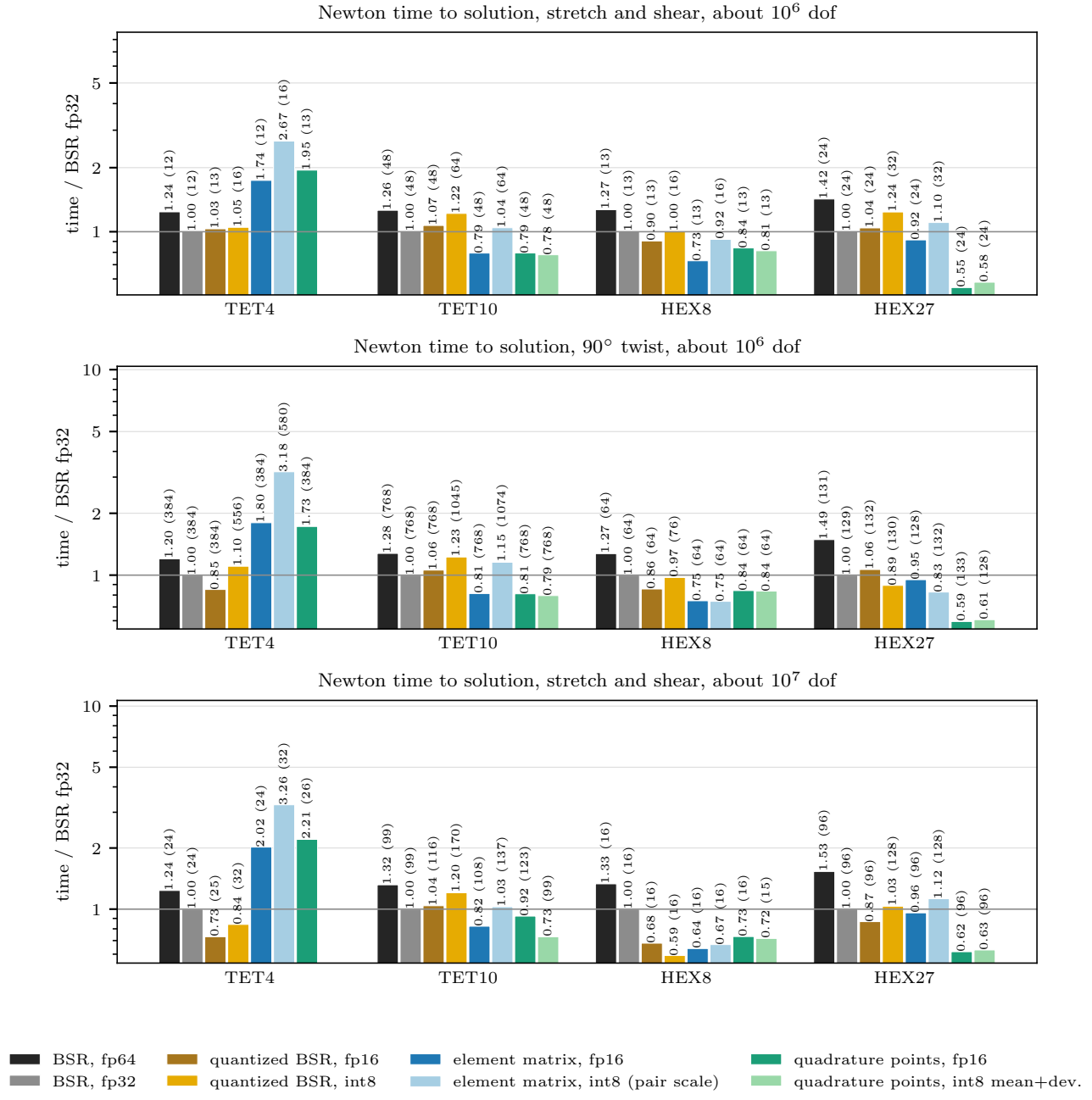

Newton time to solution, stretch and shear, about 10^6 dof
time / BSR fp32
1.24 (12)
1.00 (12)
1.03 (13)
1.05 (16)
1.74 (12)
2.67 (16)
1.95 (13)
1.26 (48)
1.00 (48)
1.07 (48)
1.22 (64)
0.79 (48)
1.04 (64)
0.79 (48)
0.78 (48)
1.27 (13)
1.00 (13)
0.90 (13)
1.00 (16)
0.73 (13)
0.92 (16)
0.84 (13)
0.81 (13)
1.42 (24)
1.00 (24)
1.04 (24)
1.24 (32)
0.92 (24)
1.10 (32)
0.55 (24)
0.58 (24)
TET4
TET10
HEX8
HEX27
Newton time to solution, 90° twist, about 10^6 dof
time / BSR fp32
1.20 (384)
1.00 (384)
0.85 (384)
1.10 (556)
1.80 (384)
3.18 (580)
1.73 (384)
1.28 (768)
1.00 (768)
1.06 (768)
1.23 (1045)
0.81 (768)
1.15 (1074)
0.81 (768)
0.79 (768)
1.27 (64)
1.00 (64)
0.86 (64)
0.97 (76)
0.75 (64)
0.75 (64)
0.84 (64)
0.84 (64)
1.49 (131)
1.00 (129)
1.06 (132)
0.89 (130)
0.95 (128)
0.83 (132)
0.59 (133)
0.61 (128)
TET4
TET10
HEX8
HEX27
Newton time to solution, stretch and shear, about 10^7 dof
time / BSR fp32
1.24 (24)
1.00 (24)
0.73 (25)
0.84 (32)
2.02 (24)
3.26 (32)
2.21 (26)
1.32 (99)
1.00 (99)
1.04 (116)
1.20 (170)
0.82 (108)
1.03 (137)
0.92 (123)
0.73 (99)
1.33 (16)
1.00 (16)
0.68 (16)
0.59 (16)
0.64 (16)
0.67 (16)
0.73 (16)
0.72 (15)
1.53 (96)
1.00 (96)
0.87 (96)
1.03 (128)
0.96 (96)
1.12 (128)
0.62 (96)
0.63 (96)
TET4
TET10
HEX8
HEX27
BSR, fp64
BSR, fp32
quantized BSR, fp16
quantized BSR, int8
element matrix, fp16
element matrix, int8 (pair scale)
quadrature points, fp16
quadrature points, int8 mean+dev.